PharmOS

Community pharmacy, simplified.

Product Features, Personas, Scenarios,
User Stories, and Feasibility Prototype

Part One - Product Features

1. Features concluded from the product vision

ID	Feature	Description
F1	**Inventory and Expiry Control**	Track medicines, quantity, batch and expiry data, low-stock thresholds, and barcode-based lookup. Surface low-stock and near-expiry items immediately.
F2	**Fast Sales / POS**	Create a sale by barcode scan or search, calculate totals, record the transaction, print a receipt, and deduct inventory automatically.
F3	**Customer and Prescription Log**	Maintain lightweight customer profiles with searchable prescription and purchase history. This remains a record-management feature, not a clinical diagnosis system.
F4	**Smart Stock Assistant**	Use recent sales history and current stock to identify likely shortages and suggest reorder quantities. Suggestions are advisory and require staff review.
F5	**Dashboard and Reports**	Present only actionable information: low stock, near-expiry items, sales activity, stock totals, and basic historical reports.

Scope protection

Insurance adjudication, external e-prescribing networks, clinical interaction diagnosis, delivery routing, patient mobile apps, and multi-store management are excluded from the first development stages. Security, backups, user permissions, and an audit trail remain baseline system requirements rather than headline features.

2. Personas

The personas represent the primary users described: pharmacists and pharmacy staff working at a desktop counter with keyboard, mouse, barcode scanner, and receipt-printer access.

Assigned	Persona	Role and context	Goals / pain points
Hassan	**Maya - Owner-Pharmacist Age 41**	Runs the pharmacy, purchases stock, checks performance, and still works at the counter.	Goals: Avoid stockouts and expiry waste; keep purchasing under control; understand the pharmacy quickly. Pain points: Limited time, uncertain reorder quantities, and frustration with broad systems that bury common tasks.
Dana	**Karim - Staff Pharmacist Age 29**	Handles prescriptions, counter sales, customer records, and frequent interruptions.	Goals: Complete sales quickly; find customer and stock information immediately; avoid mistakes during rush periods. Pain points: Too many clicks, slow searches, and having to leave the sales workflow to check inventory.
Nicolas	**Rania - Pharmacy Technician Age 24**	Receives deliveries, scans items, enters batch/expiry information, and organizes stock.	Goals: Enter incoming stock accurately and quickly; catch bad or near-expiry deliveries before shelving. Pain points: Repetitive data entry, barcode mismatches, duplicate batches, and manual expiry checking.

3. Scenarios of Product Use

Scenario A - Monday inventory planning

Maya opens PharmOS before placing the week's supplier order. The dashboard shows three high-risk stock items. She opens Smart Stock, reviews the recommendation for Amoxicillin, checks the recent sales rate and supplier lead time, changes the suggested quantity from 24 to 20 because she expects a quiet week, and adds it to a draft reorder list. Nothing is sent automatically.

Features exercised: F4 Smart Stock Assistant, F6 Reorder Review and Draft Order

Scenario B - Rush-hour counter sale

Karim scans two products while several customers are waiting. PharmOS immediately finds each item, displays the price and available stock, and keeps the entire transaction inside the POS screen. He optionally links the sale to an existing customer, completes payment, prints the receipt, and the inventory count is reduced automatically. If a product crosses its low-stock threshold, the dashboard alert updates.

Features exercised: F1 Inventory and Expiry Control, F2 Fast Sales / POS, F3 Customer and Prescription Log, F5 Dashboard and Reports

Scenario C - Receiving a supplier delivery

Rania receives a delivery and switches to Stock Receiving mode. She scans each package, enters the received quantity, batch number, and expiry date, and confirms the line. PharmOS warns her when an expiry date is unusually close or when a batch already exists, then updates inventory only after confirmation.

Features exercised: F1 Inventory and Expiry Control, F7 Stock Receiving Mode

Features added or extended by the scenarios

ID	Scenario-derived feature	Why it was added
F6	**Reorder Review and Draft Order**	Converts Smart Stock recommendations into an editable draft reorder list. Staff can change quantities, remove items, and confirm the list. PharmOS never submits an order automatically.
F7	**Stock Receiving Mode**	A focused receiving workflow for barcode scan, quantity, batch, and expiry capture. It validates incoming data and warns about duplicate batches or unusually close expiry dates before inventory is updated.

4. User Stories

Each feature is described using user stories in the form: As a [user], I want [goal], so that [benefit]. The stories are intentionally small enough to be implemented and tested independently.

F1 - Inventory and Expiry Control

- As a pharmacy technician, I want to scan a medicine and enter its quantity, batch, and expiry, so that inventory records stay accurate.
- As a pharmacist, I want low-stock and near-expiry items to be flagged automatically, so I can act before stockout or expiry loss occurs.

F2 - Fast Sales / POS

- As a staff pharmacist, I want to add a product by barcode scan or search, so that I can build a sale quickly.
- As a staff pharmacist, I want completing a sale to record the transaction, print a receipt, and deduct stock automatically, so that I do not repeat the same work in multiple screens.

F3 - Customer and Prescription Log

- As a pharmacist, I want to search for a customer and see prescription and purchase history, so that I can review prior records quickly.
- As a pharmacist, I want to associate a prescription or sale with a customer when needed, so that the history remains searchable later.

F4 - Smart Stock Assistant

- As an owner-pharmacist, I want shortage-risk items ranked by urgency, so that I know what deserves attention first.
- As an owner-pharmacist, I want each suggestion to show the sales rate, stock on hand, supplier lead time, and estimated days of cover, so that I can understand why the system made the recommendation.
- As an owner-pharmacist, I want the assistant to remain advisory, so that I retain control over purchasing decisions.

F5 - Dashboard and Reports

- As a pharmacy owner, I want the homepage to show low stock, expiry risk, recent sales, and stock totals, so that I can understand the pharmacy's current state quickly.
- As a staff member, I want clicking an alert to open the relevant filtered inventory view, so that I do not have to search for the item manually.

F6 - Reorder Review and Draft Order

- As an owner-pharmacist, I want to edit a suggested reorder quantity before adding it to a draft list, so that the order reflects my judgment.
- As an owner-pharmacist, I want to review the complete draft before confirmation, so that no recommendation becomes an order without explicit approval.

F7 - Stock Receiving Mode

- As a pharmacy technician, I want to scan incoming products and enter batch and expiry data in one receiving screen, so that delivery entry is fast and consistent.
- As a pharmacy technician, I want warnings for duplicate batches or unusually close expiry dates, so that I can correct questionable delivery data before updating inventory.

Part Two - Feasibility Prototype

Three-screen Figma walkthrough

The feasibility prototype is presented as a three-screen interaction path. The first screen is the unchanged PharmOS. The next two screens dive into the new Smart Stock idea without redesigning the original homepage.

Screen	Starting point	User action	Destination
1 - Homepage	Home dashboard	Click View All in Low Stock Alerts	Inventory - Attention Needed
2 - Inventory attention	Filtered low-stock / near-expiry list	Click Analyze with Smart Stock on Amoxicillin	Smart Stock recommendation
3 - Smart Stock	Recommendation detail	Review/edit quantity, then create a draft order	Human-confirmed reorder draft

Screen 1 - Home Dashboard

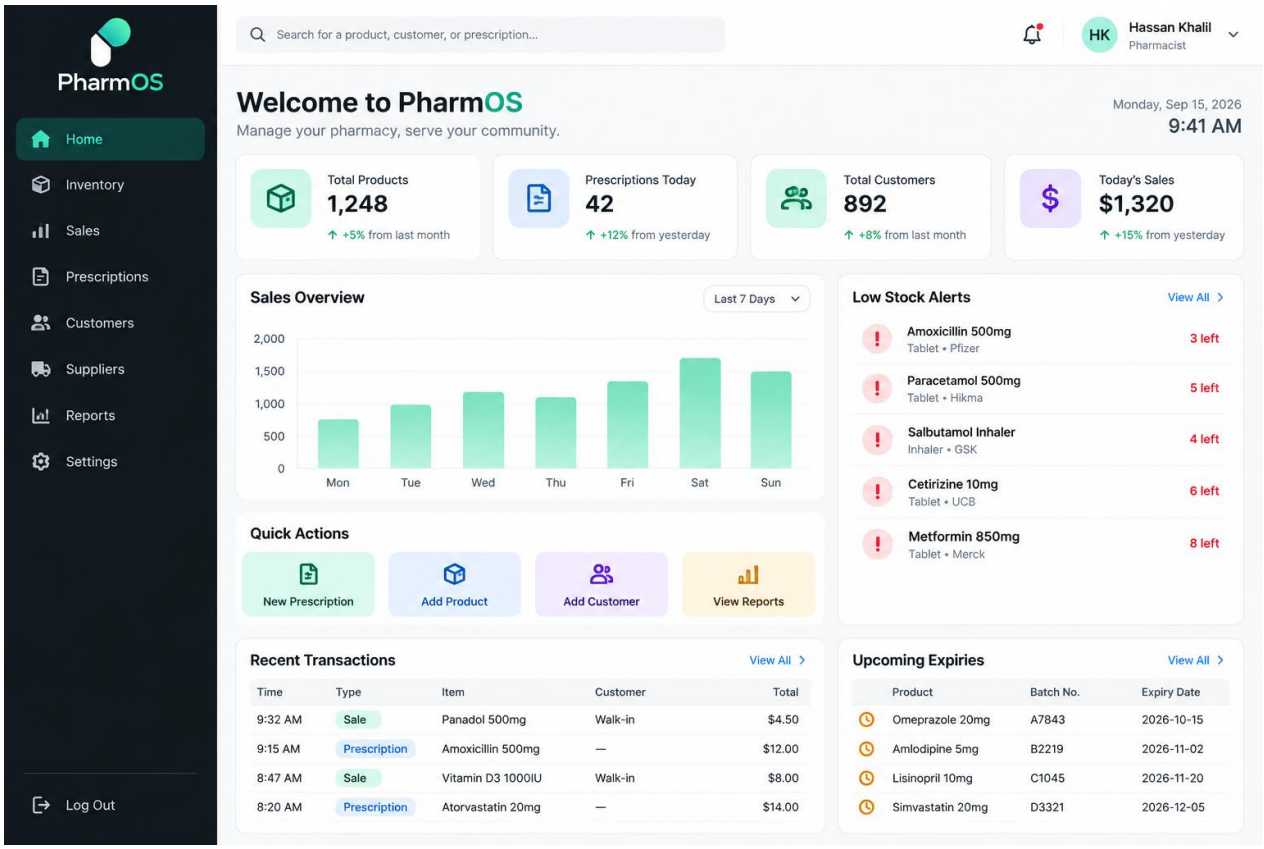


Figure 1. PharmOS homepage

Figma: Open Screen 1 - Home Dashboard

Prototype action: click View All in the Low Stock Alerts card.

Screen 2 - Inventory: Attention Needed

The alert opens a focused inventory view instead of the full catalog. Low-stock and near-expiry items are prioritized, and the highest-risk item is selected. The important new action is Analyze with Smart Stock.

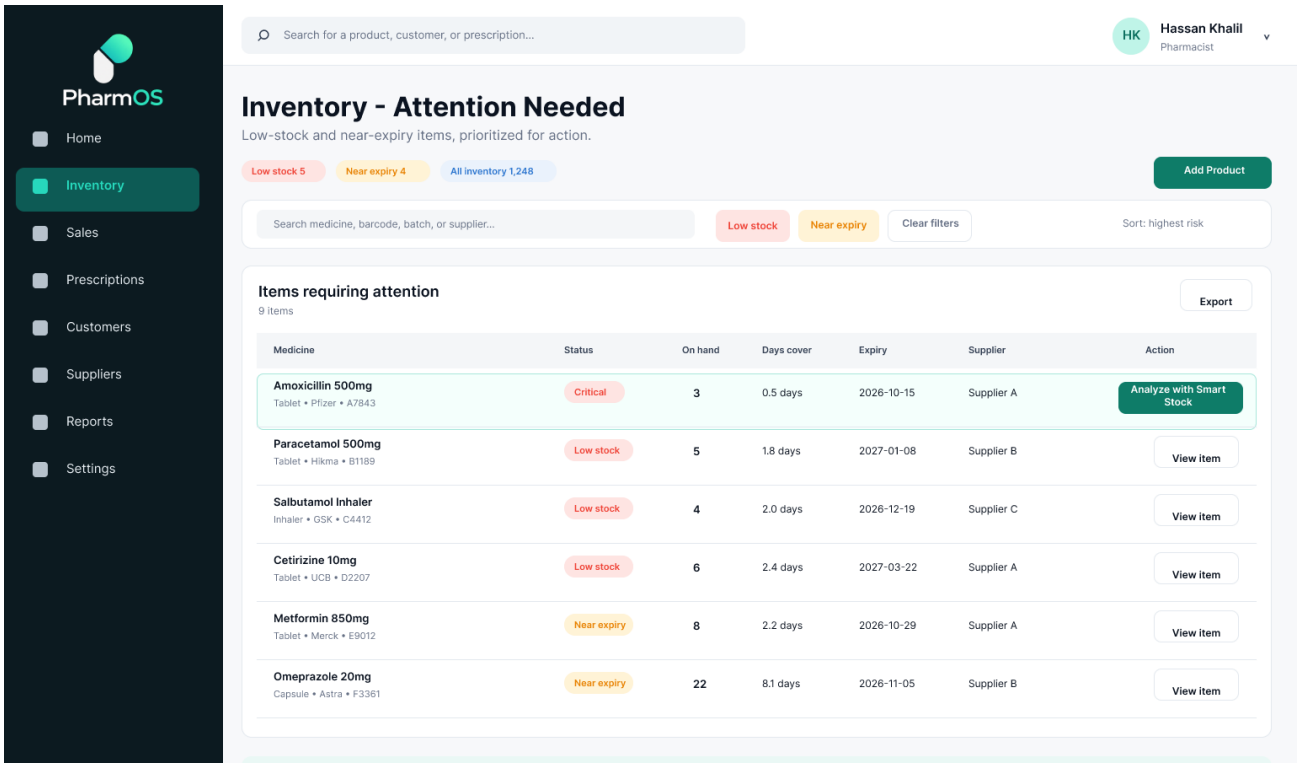


Figure 2. Filtered inventory attention view - sample data.

Figma: Open Screen 2 - Inventory Attention **Prototype action: click Analyze with Smart Stock for Amoxicillin 500mg.**

Screen 3 - Smart Stock Recommendation

The final screen explains why the selected medicine is at risk, exposes the data used in the recommendation, proposes an editable reorder quantity, and keeps the final action under pharmacist control. No order is submitted automatically.

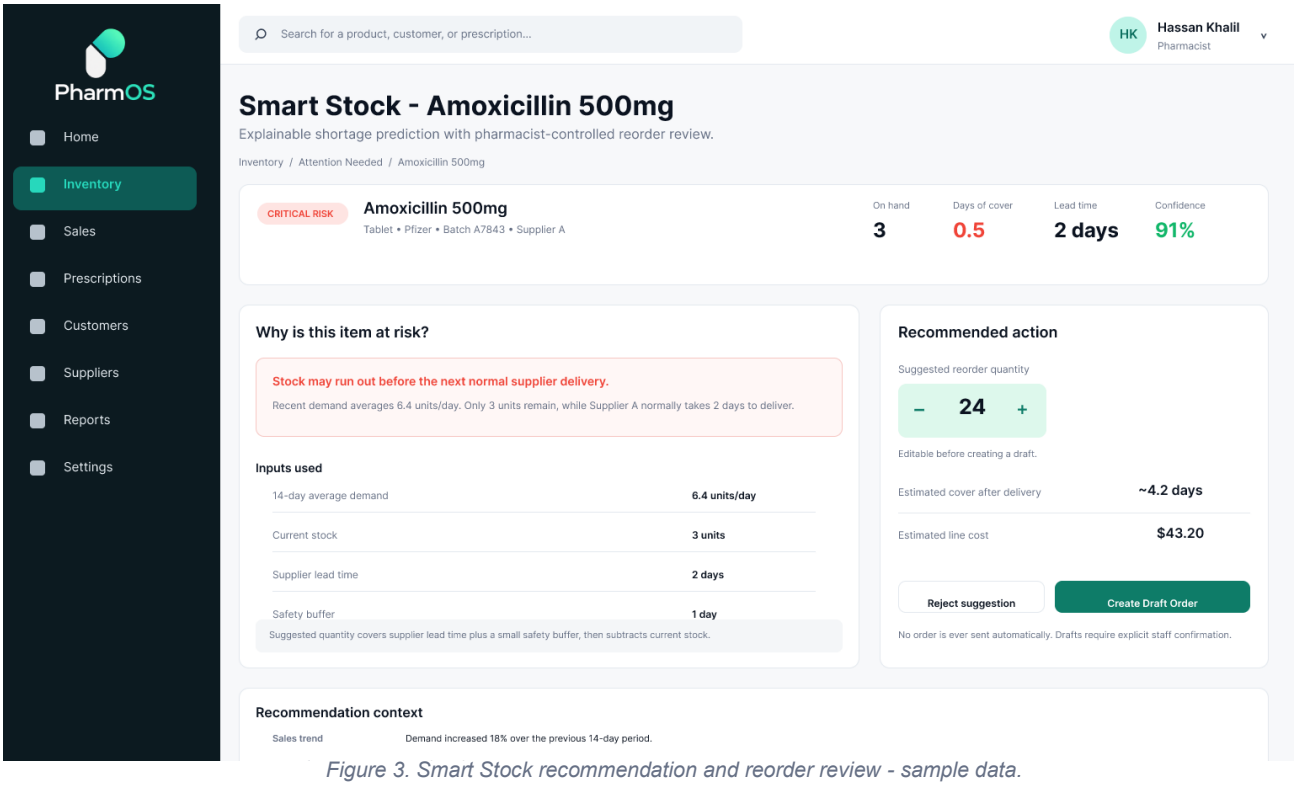


Figure 3. Smart Stock recommendation and reorder review - sample data.

Figma: Open Screen 3 - Smart Stock Recommendation

What the three-screen prototype demonstrates

- Continuity: the new feature starts from the existing homepage rather than requiring a redesigned interface.
- Progressive disclosure: homepage shows problem, inventory screen narrows problem, and Smart Stock explains the proposed action.
- Explainability: the recommendation exposes demand, stock, lead time, days of cover, and the suggested quantity.
- Human-controlled automation: the user can edit or reject the recommendation, and the system creates only a draft order for review.

Feasibility Demonstration

Initial implementation approach

The first version does not require a complex machine-learning model. It can be implemented with stored sales and inventory data, then upgraded later if more advanced forecasting proves useful. This reduces technical risk while still testing whether the workflow provides value.

Simple feasibility logic

Average daily demand = units sold in recent window / number of days

Days of cover = current stock / average daily demand

Suggested reorder = max(0, forecast demand during lead time + safety buffer - current stock)

Example: if an item averages roughly 6.4 units/day, only 3 units remain, and the supplier normally takes two days, the assistant can flag the item as critical and propose a quantity large enough to cover the replenishment period plus a small safety buffer. The pharmacist can then edit or reject that value.

Comparison with competing products

The comparison is deliberately narrow. PioneerRx, Liberty, PrimeRx, and BestRx are broader commercial systems; PharmOS is not claiming to exceed them overall. The question is whether the proposed new workflow is a better fit for the specific PharmOS target: a small community pharmacy that values clarity, low complexity, and explicit staff control.

Competitor capability	Comparable capability	PharmOS design advantage for its target user
PioneerRx	Usage-based and recommended ordering; system-generated reorder points are available, and PioneerRx also connects to predictive inventory services.	PharmOS places the explanation next to each recommendation and keeps the approval step visible. The improvement is transparency and a smaller decision surface, not broader functionality.
Liberty PharmacyOne	Inventory control includes a Want Book, automatic order creation, multiple vendors, and integrated wholesaler ordering.	PharmOS adds a compact explanation of why an item is risky and separates recommendation from confirmation. This favors users who want to understand and manually approve the proposed action.
PrimeRx	Supports inventory forecasting and can automatically reorder when preset levels are reached.	PharmOS deliberately avoids automatic submission. It is a better fit when the design goal is human-controlled automation and every reorder must remain reviewable.
BestRx	Provides inventory/usage reports, reorder points, purchase-order history, and wholesaler interfaces.	PharmOS turns recent usage into a forward-looking, item-level shortage suggestion with estimated days of cover and an editable proposed quantity directly in the workflow.

Prototype conclusion

The three-screen feasibility prototype supports the product vision without expanding scope. It begins with the unchanged homepage, narrows an inventory alert into a prioritized item view, and then demonstrates the new explainable Smart Stock recommendation. The recommendation can be computed from data PharmOS already stores, the user can understand its basis, and the final purchasing decision remains under staff control.

References

[1] PharmOS - Product Vision and Rapid Prototyping.

[2] PioneerRx - Pharmacy Software: https://www.pioneerrx.com/pharmacy-software

[3] Liberty Software - PharmacyOne: https://libertysoftware.com/pharmacy/

[4] PrimeRx - Pharmacy inventory forecasting and management: https://www.primerx.io/blog/optimizing-inventory-for-flu-season-primerx-can-help

[5] BestRx - Pharmacy inventory management: https://www.bestrx.com/blog/3-ways-bestrx-helps-you-manage-your-pharmacys-inventory